The two ends of the critical configuration

Human Compression project: a component of the L019 recovery

We retain the standing hypotheses and strict smaller-degree induction of the acquired first-cut component. Thus the full fractured natural symmetric string C-marking consists of an even degree-six rank-three primitive A_5 factor E on Y, followed by

$$P = \langle t_0, \dots, t_{s-1} \rangle = \mathrm{Sym}(X), \qquad |X| = m, \quad s = m/2 + 1, \quad X \cap Y = \{a\}.$$

The common point is an endpoint anchor in each factor; m is even. The full degree is $n = m + 5$ and its rank is $m/2+4$. A sufficient split has the precise meaning fixed in the acquired component: it is pure, or has a trivial internal factor and two odd actual one-point attaching groups. The induction applies only to smaller degrees satisfying the strict inequality $2u > d + 2$.

We may assume the least unique crossing of P has index zero; the acquired first-cut theorem has disposed of every positive index. Write

$$t_0 = (a\ b)\eta, \qquad T = \langle t_1, \dots, t_{s-1} \rangle.$$

Its perfect split has singleton side $\{a\}$ and complementary side $X \setminus \{a\}$. The actual-wing bound, since $2\,\mathrm{rank}(T) = m > m - 1$, gives $T = \mathrm{Sym}(X \setminus \{a\})$.

Lemma 1. *Either the full marking has an internal sufficient split, or T has a least unique crossing at an index $k \geq 2$ in its own order. It is a perfect split of T and inherits to the full marking at the original P index $j = k + 1$. If A', B are its two T-omission sides, with $b \in A'$, then*

$$|A'| \geq 2k + 1, \quad \eta \text{ is supported on } A', \quad a_0 := |\{a\} \cup A'| \geq 2j. \tag{1}$$

If $a_0 > 2j$, the full marking has a sufficient split.

Proof. The tail is fractured, inherits IP as an actual supported subgroup, and has endpoint anchor b. Its degree is $m-1$ and twice its rank is m, so the acquired anchored first-split theorem applies. It supplies the perfect split and $2k \leq |A'| - 1$. If $k = 0$, the adjacent inheritance lemma applies with tail rank $m/2$ and degree $m - 1$: its strict condition is $m > m - 2$. It forces $\eta = 1$. The original mark t_0 is then pure and is internal in the full marking, because three E marks precede it and the nonempty tail follows it. This is sufficient. If $k = 1$, the one-mark left wing of T is transitive of degree two, contrary to $|A'| \geq 3$.

For $k \geq 2$, nonadjacent inheritance at the original index zero applies to T's split. In its actual support proof, t_0 fixes B pointwise, so η is supported on A'. The new left side of P is $\{a\} \cup A'$, giving (1). The same split in the full marking has sides $Y \cup A'$ and B: the E factor attaches only at a, and the split mark fixes Y.

Put $\delta = a_0 - 2j$. When $\delta \geq 1$, the acquired positive-defect proof applies verbatim, although j need not be the least unique index of P. To make this interface explicit, that proof after its first lemma uses only a perfect split, $j \geq 2$, and $a_0 \geq 2j + 1$. These give right-wing rank q with $2q = |B| + \delta > |B|$, hence an actual symmetric right wing. The remaining faithful-augmentation and parity cases use exactly these quantities and the full strict parent. They therefore provide the same sufficient split under the same smaller-degree induction. □

Theorem 2. *If the full marking has no internal sufficient split, then for some $k \geq 2$ its labels and actual supports have the form*

$$E,\ q_-,\ h_0, \ldots, h_{k-1},\ q_+,\ E'. \tag{2}$$

There are pairwise disjoint sets Y, W, Z, of sizes $6, 2k+1, 6$, on which $E, M := \langle h_0, \ldots, h_{k-1} \rangle, E'$ are respectively supported and transitive. Both E, E' are even primitive degree-six rank-three A_5 factors. For $a \in Y$, $d \in Z$ and $b, c \in W$,

$$q_- = (a\ b)\eta, \qquad q_+ = (c\ d)\alpha, \tag{3}$$

where η, α are nonidentity involutions supported on W, fixing b, c respectively. The domain has size $2k + 13$ and the rank is $k + 8$. Moreover, as actual supported groups,

$$\begin{aligned} M &= \mathrm{Sym}(W), \\ \langle q_-, M \rangle &= \mathrm{Sym}(W \cup \{a\}), \qquad \langle M, q_+ \rangle = \mathrm{Sym}(W \cup \{d\}), \\ \langle q_-, M, q_+ \rangle &= \mathrm{Sym}(W \cup \{a, d\}). \end{aligned} \tag{4}$$

The marking of M is fractured, has endpoint anchors b, c, and has at least two eligible edges of every colour.

Proof. By Lemma 1, only $\delta = 0$ remains. Put $W = A'$, so $|W| = 2k+1$, $j = k+1$, and $a_0 = 2j$. Set $h_i = t_{i+1}$ for $0 \leq i < k$, and $q_- = t_0$, $q_+ = t_j$. The left full wing

$$F = \langle E, t_0, \ldots, t_{j-1} \rangle$$

has degree $f = a_0 + 5$ and rank $j + 3$, with $2(j + 3) = f + 1$. The actual-wing bound forces $F = \mathrm{Sym}(Y \cup W)$. Write the right wing as R, with degree $z = |B|$ and rank q. The critical rank identity gives $2q = z$. In particular z is positive and even. Write the split mark

$$t_j = (c\ d)\alpha\beta$$

with internal factors on the two full sides. Its left factor fixes Y, since t_j does, and is supported on W.

If both factors are nonidentity, faithful augmentation with symmetric wing F gives a proper primitive faithful C-marking of degree $z + 1$ and rank $q + 1$. Its rank is strictly above half its degree. The acquired primitive classification allows proper exceptions only in degrees six and ten, whereas $z+1$ is odd. This is impossible, including the small odd degrees; natural alternating and symmetric groups are excluded by the same augmentation theorem.

Suppose $\alpha = 1$. The attaching groups are now F and $J = \langle t_j, R \rangle$, meeting at c. Since F is odd, an odd J gives a sufficient split. If J is even, the acquired even-attachment theorem gives either $2\,\mathrm{rank}(F) > f + 2$ or the exceptional right degree $z = 5$. The former contradicts $2\,\mathrm{rank}(F) = f + 1$ and the latter contradicts the evenness of z. This case is disposed of.

Consequently $\beta = 1$. The group $H = \langle F, t_j \rangle$ is $\mathrm{Sym}(Y \cup W \cup \{d\})$: its symmetric subgroup F contains α, so it also contains the single crossing transposition and hence generates that full symmetric group. It meets R only at d. If R is odd, this is a sufficient split. Otherwise R is even, fractured, anchored at d, and has rank $z/2$. The acquired anchored-even theorem excludes $z \geq 12$, and the small-even theorem applies for $z = 6, 8, 10$ with the actual odd external group H. Degrees two and four cannot be even transitive C-markings of the required ranks: degree two has no even involution, while every nonidentity even involution of degree four is a fixed-point-free double transposition, incompatible with the anchor fixed by the second mark. Thus the only remaining case is $z = 6$, $q = 3$, and $R = E'$ is the stated primitive A_5. This proves (2)–(3). Nonidentity of both remaining internal factors follows because otherwise one of the two connecting marks would already be a pure internal split.

It remains to identify the middle groups without projecting any C-marking. The original P and T are the actual groups

$$P = \mathrm{Sym}(\{a\} \cup W \cup Z), \qquad T = \mathrm{Sym}(W \cup Z).$$

We have just proved $F = \mathrm{Sym}(Y \cup W)$ and $H = \mathrm{Sym}(Y \cup W \cup \{d\})$. The intersections of supported symmetric groups are the symmetric groups on their set intersections. The original IP, applied to their exact label sets, therefore gives

$$\begin{aligned} F \cap T &= M = \mathrm{Sym}(W), & F \cap P &= \langle q_-, M \rangle = \mathrm{Sym}(W \cup \{a\}), \\ H \cap T &= \langle M, q_+ \rangle = \mathrm{Sym}(W \cup \{d\}), & H \cap P &= \langle q_-, M, q_+ \rangle = \mathrm{Sym}(W \cup \{a, d\}). \end{aligned}$$

These are literal subgroup intersections in the given ambient group; no quotient-IP assertion or new finite exception test enters.

The unique crossings in (3) and the string relations show that all middle marks except h_0 fix b, and all except h_{k-1} fix c. Their transitivity makes the respective end marks move these points. Thus b, c are endpoint anchors. They are distinct because $k \geq 2$: otherwise every middle mark would fix their common point. Fracture descends from T, since a transitive middle omission, together with q_+ and E', would give a transitive omission of T.

Finally k was the least unique crossing index in T. Every preceding colour therefore has at least two eligible edges in T. Every edge of those marks lies in W. Their eligibility persists upon passing to the smaller omission subgroup of M. This proves the final assertion and completes the reduction. □

Attribution and remaining task. The source extension problem and the acquired classifications remain attributed to Cameron, Fernandes, Leemans and Mixer through their individual recovered statements. This deduction uses the new first-cut density and actual intersections to isolate the two-ended configuration. In particular, (4) avoids a further primitive classification or finite small-degree calculation. The remaining middle marking is symmetric of degree $2k + 1$ and rank k, with two endpoint anchors and paired eligible edges. It has not been classified or excluded here. Neither the full extension assertion nor its inverse counting correspondence is claimed. There is no whole-original, conjecture, Lean or measurement credit in this component.